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Polyfluorinated tetraoxacalixarenes and bicyclooxacalixarenes. Interaction of pentafluorobenzonitrile with resorcinol, orcinol and tetrafluororesorcinol

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ABSTRACT

Successive interaction of pentafluorobenzonitrile with resorcinol, orcinol and tetrafluororesorcinol in Et_3N-CH_3CN system gives fluorinated tetraoxacalix[4]arenes ABAB and ABAC type in a good yield. Further reaction of obtained products with new additional equivalent of tetrafluororesorcinol or orcinol leads to the formation of the fluorine-containing bicyclooxacalixarenes. The ESI-MS method shows the possibility of fluorinated tetraoxacalixarenes and bicyclooxacalixarenes complexes formation with metal cations.

1. Introduction

Oxacalixarenes are one of the platforms in supramolecular chemistry suitable for the study of intermolecular guest-host interactions [1]. Several of oxacalixarene-based chemical sensors for organic compounds [1,2], cations [1,3] and anions [4] are known. It was proposed to use tetraoxacalixarenes in high-performance liquid chromatography, which allows analyzing impurities such as benzpyrene and melamine in food products [5]. The possibility of using oxacalixarenes as transporters of ion pairs through cell membranes has been shown [6]. In parallel with the synthesis of oxacalixarenes, methods for their modification and construction of more complex molecular structures based on them were also developed [1]. Such close and almost parallel location of "nucleophilic" aromatic cycles in tetraoxacalixarenes allows to connect them by various linkers with the formation of 3D-molecular structures of the type oxacalixcrown ethers [1,7] or bicyclooxacalixarenes [1,8–11]. The use of phloroglucinol as nucleophilic component in the synthesis allows obtaining bicyclooxacalixarenes in one step [9]. The electrophilic components in these reactions are 1,5-difluoro-2,4-dinitrobenzene [9] or various 1,3-dihaloheteroaromatic compounds [9–11]. Bicyclooxacalixarene synthesized from 2,4,6-trichloro-1,3,5-triazine and phloroglucinol like oxacalixarene forms complexes with halogen anions [10]. On the basis of the same compound, an even more complex 3D-molecular structure, called a molecular barrel, was obtained [12]. Interaction of pyrazine-based bicyclooxacalixarene with salts of Ag^+, Cu^{2+} and Zn^{2+} leads to one- and two-dimensional metallosupramolecular polymers [13].

The data of the X-ray analysis of polyfluorinated tetraoxacalixarenes [14,15] showed that opposite located fluorinated aromatic cycles are quite close and form a sufficiently acute dihedral angle (40-49°), probably due to the repulsion of fluorine atoms in the internal cycle of the macromolecule. This location of polyfluorinated aromatic nuclei suggests the possibility of their binding to each other by linkers using aromatic nucleophilic substitution reactions. A key necessary condition for this is the presence of another one activated halogen atom in "electrophilic" component of molecule. The possibility of successive substitution of 3 fluorine atoms in pentafluorobenzonitrile by various aryloxy groups is known [16]. Substitution of several fluorine atoms in pentafluorobenzonitrile by N-nucleophiles is also known [17]. Recently, we used pentafluorobenzonitrile in the synthesis of perfluorinated A_3B tetraoxacalixarenes [14]. In this work, we investigated fragment approach to the formation of polyfluorinated oxacalixarenes and bicyclooxacalixarenes in the reaction of pentafluorobenzonitrile with resorcinol, orcinol and tetrafluororesorcinol.

2. Results and discussion

2.1. Synthesis of polyfluorinated tetraoxacalix[4]arenes and bicyclooxacalixarenes

A fragment approach to the synthesis of fluorinated tetraoxacalixarenes was used earlier in the reaction of 4-substituted tetrafluoropyridine with resorcinol and orcinol [4a,18]. Previously, we have shown the possibility of synthesis of polyfluorinated

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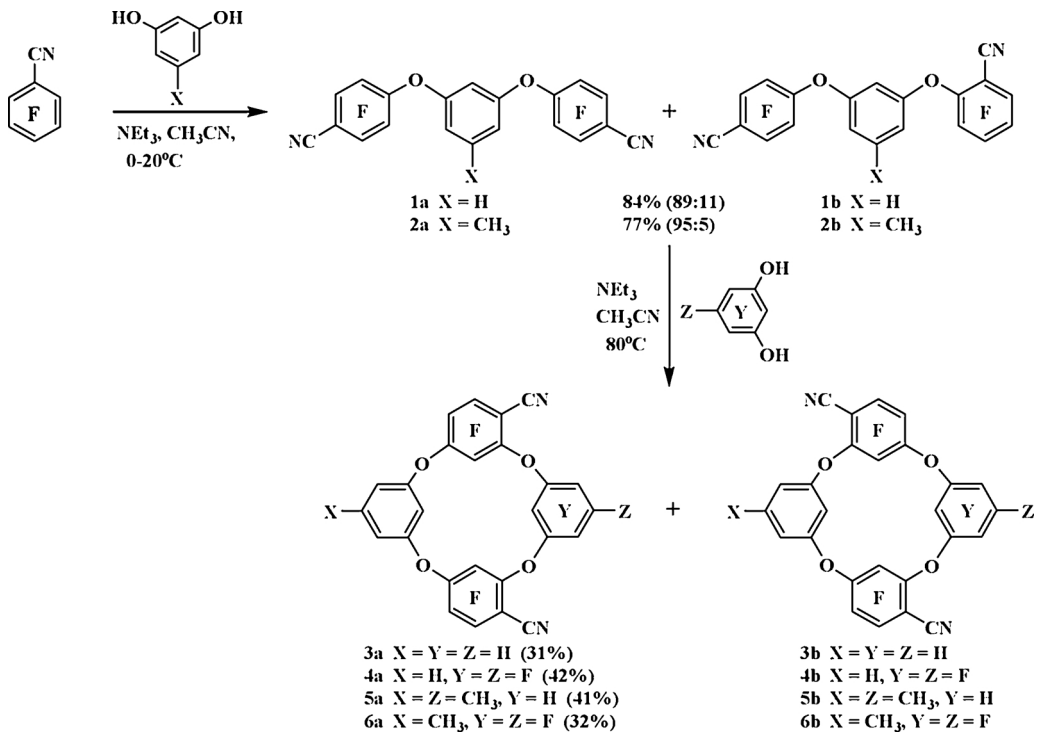


Scheme 1. Successive interaction of pentafluorobenzonitrile with resorcinol, orcinol and tetrafluororesorcinol.

tetraoxacalixarenes ABAC type by the consecutive interaction of perfluoro-m-xylene with resorcinol and tetrafluororesorcinol [15]. Similarly in the reaction of pentafluorobenzonitrile with resorcinol, orcinol and tetrafluororesorcinol in NEt_3-CH_3CN system triphenyl ethers **1a, 2a** and tetraoxacalixarenes **3a – 6a** with a small admixture of the corresponding isomers **1b, 2b** and **3b – 6b** were obtained (Scheme 1). Earlier, formation of a mixtures of isomers (ratio of 9-10:1) was found in the reactions of pentafluorobenzonitrile with sodium salt of pentafluorophenol, and dilithium salt of tetrafluororesorcinol [19].

Tetraoxacalixarenes **3a,b – 6a,b** can be synthesized from triphenyl ethers **1a, 2a** or by the successive interaction of 2 equivalents of pentafluorobenzonitrile with the equivalent of resorcinol (orcinol) under mild conditions and heating of the resulting mixture with another equivalent of resorcinol derivatives.

Prolonged heating of mixture tetraoxacalixarene **4a,b** (73%) and triphenyl ether **1a** (25%) with tetrafluororesorcinol in NEt_3-CH_3CN system leads to the formation of a products from which polyfluorinated bicyclooxacalixarene **7** was isolated with low yield (Scheme 2).

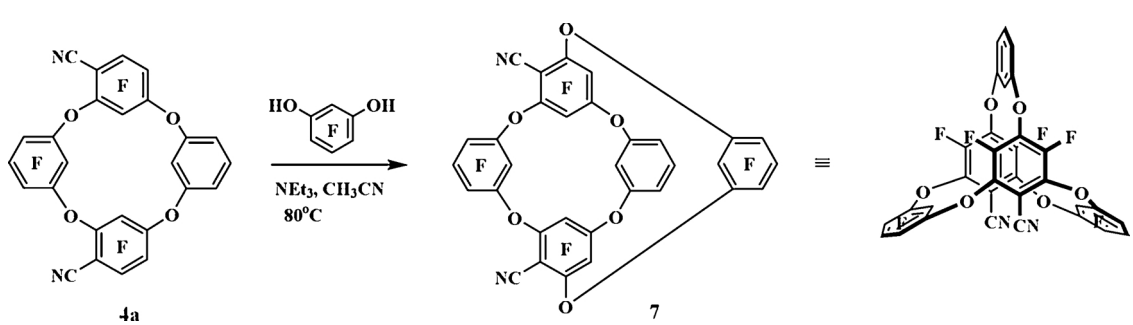
Similarly, prolonged heating of mixture tetraoxacalixarenes **5a,b** (89%) and triphenyl ether **2a** (10%) with an additional equivalent of orcinol in NEt_3-CH_3CN system gives the bicyclooxacalixarene **8** which was isolated with low yield in the mix (69:31) with initial compound **5a** (Scheme 3).

2.2. Structure of compounds

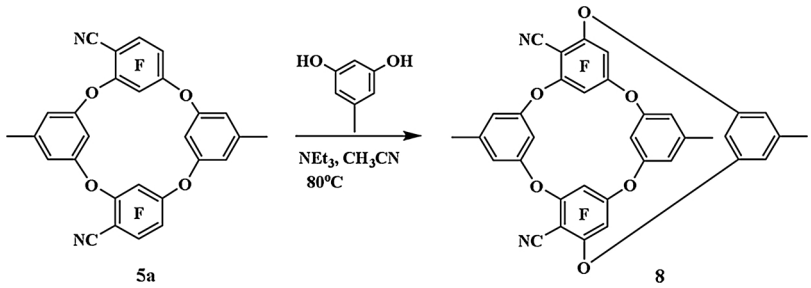
It is known, that bicyclooxacalixarenes have D_{3h}-symmetry with a parallel position of nucleophilic benzene nuclei linked by three electrophilic aromatic or heteroaromatic rings [9]. In the ^1H NMR spectra of bicyclooxacalixarenes a significant high-field chemical shift for the signals of inner-cavity protons was observed [9]. Contrarily, in the bicyclooxacalixarenes **7** and **8** two parallel electrophilic fluorinated benzonitrile fragments linked by nucleophilic resorcinol and two tetrafluororesorcinol nuclei or by three orcinol rings.

While in oxocalixarene **5a** ^1H NMR signals of inner-rim protons was observed at 5.81 and 5.96 ppm ($CDCl_3$), the formation of bicyclooxacalixarene **8** changed chemical shift of inner-cavity protons to a more higher field (4.87 (2 H), 5.02 (1 H) ppm, respectively). In turn, in the ^1H NMR spectrum, the signals of inner-rim protons in oxacalixarene **5a** also have a typical for tetraoxacalixarene structure high-field chemical shift in comparison with the triphenyl ether **2a**. In the ^1H NMR spectra of bicyclooxacalixarene **7** for the signal of inner-cavity proton also observed significant high-field chemical shift (δ 4.76 in $CDCl_3$) as compared to the one of inner-rim proton in oxacalixarenes **4a** (δ 5.95 in $CDCl_3$).

An interesting question is the possibility of fluorine atoms location inside the cavity formed by the electrophilic fluorine-containing



Scheme 2. Synthesis of bicyclooxacalixarene[5]arene **7**.



Scheme 3. Synthesis of bicyclooxacalixarene[5]arene **8**.

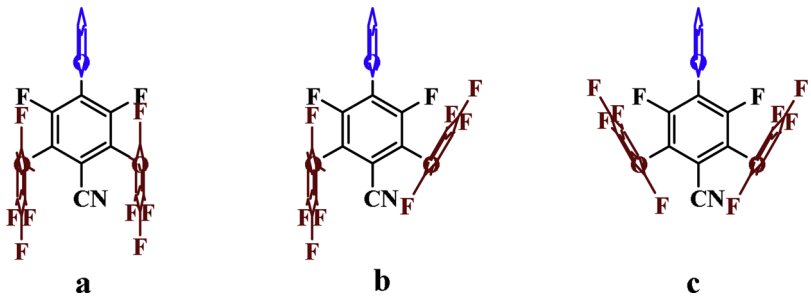


Fig. 1. Possible conformations of bicyclooxacalixarene **7**.

benzonitrile nuclei in bicyclooxacalixarene **7**. For signals of inner-cavity fluorine atoms of tetrafluororesorcinol rings of bicyclooxacalixarene **7** the high-field chemical shift in ^{19}F NMR spectra can also be expected. We observed such behavior in polyfluorinated tetraoxacalixarenes in relation to the preceding triphenyl ethers [14,15]. In contrast, inner-cavity fluorine atoms of tetrafluororesorcinol fragments of bicyclooxacalixarene **7** showed low-field chemical shift (δ -151.3 in $CDCl_3$) in comparison with the one for the inner-rim fluorine in tetraoxacalixarene **4a** (δ -154.0 in $CDCl_3$). We can assume that in the solution bicyclooxacalixarene **7** is in conformations with the tetrafluororesorcinol fragments moved aside from the cavity, as shown in Fig. 1.

One set of 4 sharp signals in the NMR spectra of compound **7** at room temperature can indicate both the realization of one symmetric conformation (**a** or **c**) and a very fast conformational equilibrium in the NMR time scale. It should also be noted that outer-rim fluorine atoms of the bicyclooxacalixarene **7**, neighboring with the bridging oxygen, showed low-field chemical shift (δ -150.3 in $CDCl_3$) in comparison with the analogous shift of fluorine atoms signals in oxacalixarene **4a** (δ -153.4 in $CDCl_3$). This low-field shift can be associated with the location of these fluorine atoms in the planes of benzonitrile rings.

According to X-ray data molecular structure of oxacalixarene **4a*0.54(CCl_4)** is close to that obtained earlier for polyfluorinated oxacalixarenes [14,15]. The dihedral angles between opposite fluorinated benzonitrile nuclei are 41.9° and 86.3° for the two other cycles. The bridging oxygen atoms are in a plane with a noticeable maximum deviation of 0.12 Å.

Bicyclooxacalixarene **7** was isolated as a crystalline complex with 3 molecules of benzene. The complex is highly unstable and loses benzene in the absence of the liquid phase to a powdery state. Bicyclooxacalixarenes are D_{3h}-symmetrical compounds with the lateral aromatic nuclei directed inside the cavity [9–11]. In the complex **7*3C_6H_6** all resorcinol and tetrafluororesorcinol fragments are moved aside from the cavity formed by benzonitrile nuclei (Fig. 2).

One of the tetrafluororesorcinol and resorcinol fragments in bicyclooxacalixarene **7** are nearly parallel (angle 3.63°). There is an interaction between the inner-cavity H and F atoms (distance 2.52 Å) of these cycles and the inner-cavity C-F bond (1.344 Å) of tetrafluororesorcinol cycle is slightly longer than the opposite C–F bond (1.334 Å), herewith inner-cavity F atom is located close to the F atoms of benzonitrile fragments (2.87 Å). Benzonitrile cycles in bicyclooxacalixarene **7** are nearly parallel: the distance between the centroids is 5.00 Å, the dihedral angle 2.46°. The CN groups are slightly deviated inward, distance N···N 4.71 Å. The bond lengths between bridging oxygen atoms and both benzonitrile and fluorinated resorcinol cycles are close (1.39 Å), whereas for benzonitrile nuclei and non-fluorinated resorcinol cycle the differences in the bond lengths of C–O are quite significant (1.37 and 1.41 Å).

Crystals of the bicyclooxacalixarene complex with benzene **7*3C_6H_6** have a layered structure with alternating layers of bicyclooxacalixarene and benzene (Fig. 3).

Benzene molecules in the complex **7*3C_6H_6** can be divided into two parts. One part of them stays in stacking interaction with tetrafluororesorcinol nuclei of bicyclooxacalixarene **7**. The distance from the benzene centroid to the planes of tetrafluororesorcinol fragments is 3.44 and 3.58 Å. The other part of the benzene molecules is oriented in the same plane with benzonitrile nuclei and interact with fluorine atoms of benzonitrile and tetrafluororesorcinol cycles.

The possibility of fluorinated tetraoxa- and bicyclooxacalixarenes to the complexation with metal ions was investigated by ESI-MS analysis, often used to test complexes formation with anions or cations [3,4,11]. The ESI(+) mass spectrum obtained for mix of oxacalixarenes **5a** with **8** and $NiCl_2$*$6H_2O$ represented in Fig. 4a.

Both oxacalixarenes **5a** and **8** give complexes with Ni^{II} ions. Ions with charge +1 are due to the interaction of one or two molecules **5a** or **8** with $[Ni^{II} + Cl]^+$: $[M_{5a} + Ni^{II} + Cl]^+$ (*m/z* 647), $[M_8 + Ni^{II} + Cl]^+$ (*m/z* 731), $[2M_{5a} + Ni^{II} + Cl]^+$ (*m/z* 1201), $[M_{5a} + M_8 + Ni^{II} + Cl]^+$ (*m/z* 1285), $[2M_8 + Ni^{II} + Cl]^+$ (*m/z* 1369). In addition, the spectrum include 4 ions with a charge of +2, formed by three molecules of **5a** or **8** and the ion Ni^{2+}: $[3M_{5a} + Ni^{II}]^{2+}$ (m/2z 860), $[2M_{5a} + M_8 + Ni^{II}]^{2+}$ (m/2z 902), $[M_{5a} + 2M_8 + Ni^{II}]^{2+}$ (m/2z 944), $[3M_8 + Ni^{II}]^{2+}$ (m/2z 986). A ESI(+) mass spectrum obtained for a mixture of oxacalixarenes **5a** and **8** with $AgNO_3$ did not showed Ag^+ complexes with oxacalixarene **5a** (Fig. 4b). Bicyclooxacalixarene **8** forms complexes with Ag^+ ion, with the inclusion 1–3 of the solvent molecules (CH_3OH): $[M_8 + Ag^I + CH_3OH]^+$ (*m/z* 777), $[2M_8 + Ag^I]^+$ (*m/z* 1383), $[2M_8 + Ag^I + CH_3OH]^+$ (*m/z* 1415), $[2M_8 + Ag^I + 2CH_3OH]^+$ (*m/z* 1447), $[2M_8 + Ag^I + 3CH_3OH]^+$ (*m/z* 1479). Bicyclooxacalixarene **7** and $NiCl_2$*$6H_2O$ did not produced detectable complexes, only $[M + Na]^+$ ion was obtained.

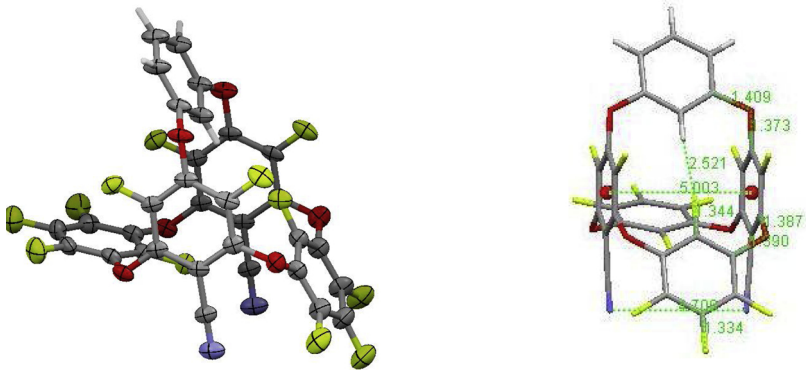


Fig. 2. Molecular structure of bicyclooxacalixarene **7**.

3. Conclusion

Macrocyclization in the reaction of pentafluorobenzonitrile with resorcinol, orcinol and tetrafluororesorcinol by the fragment (3 + 1) approach in Et_3N-CH_3CN system was investigated. It was found that the interaction of pentafluorobenzonitrile with resorcinol and orcinol in Et_3N-CH_3CN system at low temperature (0–20 °C) leads to corresponding triphenyl ethers **1a,b** and **2a,b** with a good yield. Further reaction of these triphenyl ethers **1a,b** and **2a,b** with resorcinol, orcinol and tetrafluororesorcinol under heating at 80 °C gives tetraoxacalix[4]arenes **3a,b-6a,b** ABAB or ABAC types. Prolonged heating of tetraoxacalix[4]arene **5a** with an additional equivalent of orcinol leads to bicyclooxacalixarene **8**. Polyfluorinated tetraoxacalix[4]arene ABAC-type **4a** react with an additional equivalent of tetrafluororesorcinol with the formation of the polyfluorinated bicyclooxacalixarene **7**. Chelating properties of polyfluorinated tetraoxacalixarenes and bicyclooxacalixarenes towards Ag^{I} and Ni^{II} ions were studied by ESI-MS analysis.

4. Experimental

The ^{19}F and ^{1}H NMR spectra were recorded from solutions in $CDCl_3$ on a Bruker AV300 spectrometer at 282.36 MHz and 300 MHz respectively, ^{13}C NMR spectra were recorded on a Bruker DRX-500 instrument (125.8 MHz). Chemical shifts are given in δ ppm from CCl_3F (^{19}F) and TMS (^{1}H, ^{13}C), J values in Hz; C_6F_6 (–162.9 ppm from CCl_3F), $CHCl_3$ (7.24 ppm from TMS), $CDCl_3$ (76.9 ppm from TMS) were used as internal standards. IR spectra were taken on a Bruker Vector 22 IR spectrophotometer.

The X-ray diffraction experiment for **4a*0.54(CCl_4)**, **7*3C_6H_6** was carried out on a Bruker KAPPA APEX II diffractometer with graphite monochromated MoKα radiation at 296 and 200 K accordingly. The structures were solved by direct methods using the SHELXS-97 program [20]. Refinement was carried out using the full-matrix least-squares technique using SHELXL [20]. All non-hydrogen atoms were refined anisotropically. Hydrogen atom positions were calculated geometrically and refined using the riding model. Disordered CCl_4 with partially occupied position 0.54 in **4a** and disordered benzene molecules in **7** were refined with restrictions. Crystallographic data for the structures **4a** and **7** have been deposited at the Cambridge Crystallographic Data Centre as supplementary publication no. CCDC 1898791, 1898792. Copy of the data can be obtained, free of charge, on application to CCDC, 12 Union Road, Cambridge CB21EZ, UK (fax: +44 122 3336033 or e mail: deposit@ccdc.cam.ac.uk; internet: www.ccdc.cam.ac.uk).

Gas chromatographic–mass spectrometric analysis (GC–MS) was performed on a Hewlett Packard HP 5890 Series II chromatograph coupled with an HP 5971 mass-selective detector [HP-5MS capillary column, 30 m × 0.25 mm, film thickness 0.25 μm; carrier gas helium, flow rate 1 mL/min; oven temperature programming from 50 °C (2 min) to 280 °C at a rate of 10 deg/min and finally 5 min at 280 °C; injector temperature 280 °C; ion source temperature 175 °C; electron impact, 70 eV; 1.2 scan/s, a.m.u. range 30–650].

Electrospray ionization high-resolution mass spectra (ESIHRMS) were recorded using a hybrid quadrupole/time-offlight Bruker micrOTOF-Q spectrometer using methanol as the solvent. Spectra were scanned in the *m/z* range of 100 − 3000 in positive ionization modes. Nitrogen was used as drying gas at 190 °C and at a flow rate of 4 L/min − 1. The nebulizer pressure was set to 1.0 bar. The capillary

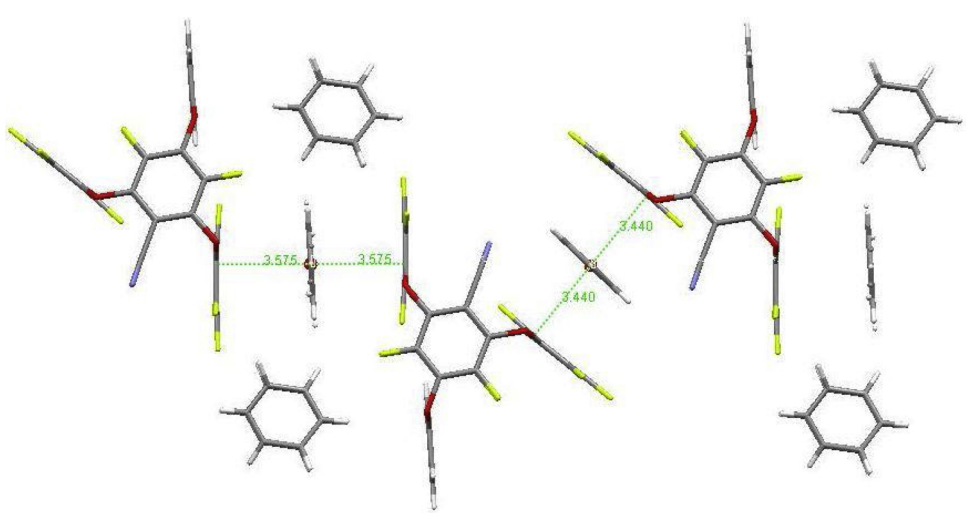


Fig. 3. Crystal packing of complex **7*3C_6H_6**.

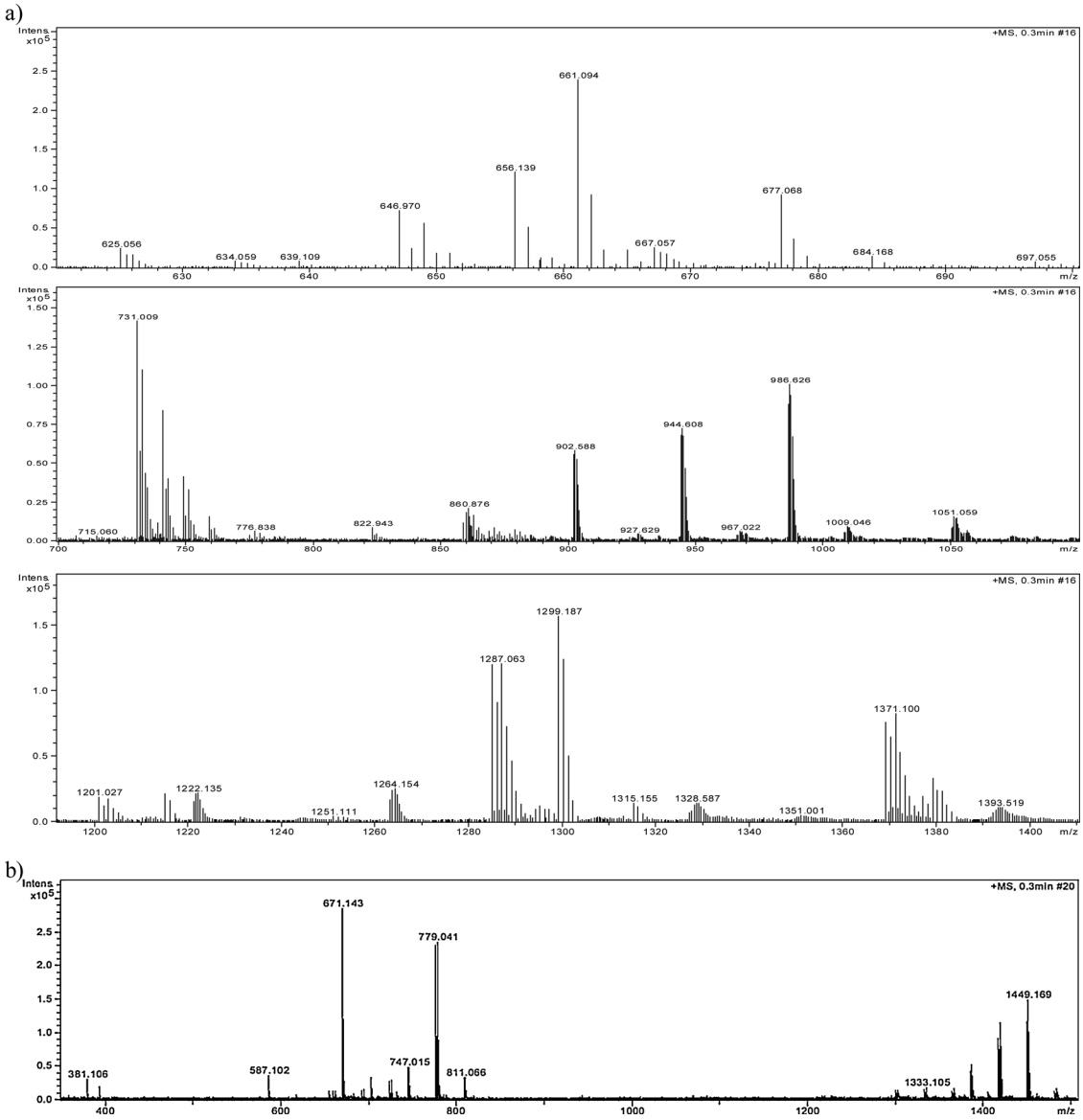


Fig. 4. ESI-MS (+) spectra for mixture of oxacalixarenes **5a** and **8** with $NiCl_2$ (a) or $AgNO_3$ (b).

voltage was set to −4.0 kV. The samples were introduced into the ESI source at the FIA mode (flow injection analysis, 2 − 3 μL at a flow rate of the solvent of 0.1 mL/min − 1). FIA was performed using an LC pump autosampler system from Agilent 1200HPLC.

The elemental compositions of tetraoxacalix[4]arene **3–6** and bicyclooxacalixarenes **7, 8** was determined from the high-resolution mass spectra which were recorded on a Thermo Scientific DFS instrument (electron impact, 70 eV).

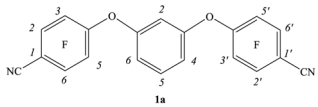
The progress of reactions was monitored by TLC on Silica gel 60 F254 plates (Merck). Silica gel (0.063–0.200 mm; Merck) was used for column chromatography.

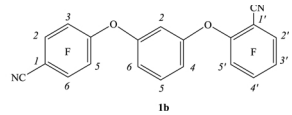
4.1. Reaction of pentafluorobenzonitrile with resorcinol and orcinol

1 Solution of 0.22 g (2 mmol) of resorcinol and 0.77 g (4 mmol) of pentafluorobenzonitrile in 50 mL of acetonitrile was added dropwise in period of 3 h with stirring into solution of triethylamine (2.40 g, 24 mmol) in 50 mL of acetonitrile, cooled to 0 °C. The mixture was stirred for 20 h at room temperature and the solvent was evaporated. Silica gel column chromatography (eluent $CCl_4/CHCl_3$ 3:1) gave 0.77 g (84%) of viscous material containing (NMR 19F, GC–MS) 85% and 9% of isomeric triphenyl ethers **1a,b**, respectively, 4% of pentafluorobenzonitrile and 2% of unidentified compounds.

2 Solution of 0.59 g (3.1 mmol) of pentafluorobenzonitrile in 2 ml was added dropwisely with stirring into solution of orcinol monohydrate (0.28 g, 2 mmol) and triethylamine (1.20 g, 12 mmol) in 10 mL of acetonitrile at 25 °C. The mixture was stirred for 2 h and the solvent was evaporated. Silica gel column chromatography (eluent $CCl_4/CHCl_3$ 3:1) gave 0.56 g (77%) of viscous material containing (NMR ^{19}F, GC–MS) 95% and 4% of isomeric triphenyl ethers **2a,b**.

4.1.1. 4,4′-[1,3-phenylenebis(oxy)]bis(tetrafluorobenzonitrile) *1a* and 2′,4-[1,3-phenylenebis(oxy)]bis(tetrafluorobenzonitrile) *1b*





Compound **1a**. Viscous liquid: ^{19}F NMR ($CCl_4 + CDCl_3$) from spectra of mixture **1a,b**: δ -151.4 (m, 4 F, F-3,5,3′,5′), -133.0 (m, 4 F, F-2,6,2′,6′). ^{1}H NMR ($CCl_4 + CDCl_3$): δ 6.81 (1H, H-2), 6.81 (d, 2H, $J_{H(4)-H(5)}$ = 8 Hz, H-4,6), 7.36 (t, 1H, $J_{H(5)-H(4,6)}$ = 8 Hz, H-5). GC–MS *m/z*: 456 (M^+), 266 (M^+ - NCC_6F_4O), 238.

Compound **1b:** ^{19}F NMR ($CCl_4 + CDCl_3$) from spectra of mixture **1a,b**: δ -157.8 (t, 1 F, $J_{F(3'3')-F(2',4')}$ = 21 Hz, F-3′), -151.4 (m, 2 F, F-3,5), -149.6 (dd, 1 F, $J_{F(5')-F(4')}$ = 20 Hz, $J_{F(5')-F(2')}$ = 9 Hz, F-5′), -144.0 (ddd, 1 F, $J_{F(4')-F(5')}$ = 21 Hz, $J_{F(4')-F(3')}$ = 20 Hz, $J_{F(4')-F(2')}$ = 6 Hz F-4′), -133.0 (m, 2 F, F-2,6), -132.0 (ddd, 1 F, $J_{F(2')-F(3')}$ = 21 Hz, $J_{F(2')-F(5')}$ = 9 Hz, $J_{F(2')-F(4')}$ = 6 Hz F-2′). GC–MS *m/z*: 456 (M^+), 238.

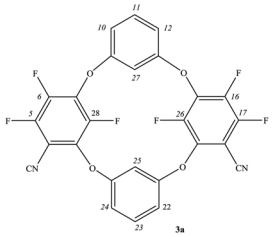
*4.1.2. 4,4′-[(5-methyl-1,3-phenylene)bis(oxy)]bis(tetrafluorobenzonitrile) **2a** and 2′,4-[(5-methyl-1,3-phenylene)bis(oxy)]bis(tetrafluorobenzonitrile) **2b***

Viscous liquid. Compound **2a**: ^{19}F NMR ($CCl_4 + CDCl_3$) from spectra of mixture **2a,b**: δ -151.6 (m, 4 F, F-3,5,3′,5′), -132.9 (m, 4 F, F-2,6,2′,6′). ^{1}H NMR ($CCl_4 + CDCl_3$): δ 2.34 (s, 3H, CH_3), 6.54 (s, 2H, H-4,5), 6.56 (s, 1H, H-2). GC–MS *m/z*: 470 (M^+), 280 (M^+ - NCC_6F_4O).

Compound **2b**:^{19}F NMR ($CCl_4 + CDCl_3$) from spectra of mixture **2a**,**b**: δ -157.7 (t, 1 F, $J_{F(3')-F(2',4')}$ = 21 Hz, F-3′), -151.6 (m, 2 F, F-3,5), -149.9 (dd, 1 F, $J_{F(5')-F(4')}$ = 20 Hz, $J_{F(5')-F(2')}$ = 9 Hz, F-5′), -144.3 (td, 1 F, $J_{F(4')-F(3,5')}$ = 20 Hz, $J_{F(4')-F(2')}$ = 6 Hz, F-4′), -132.9 (m, 2 F, F-2,6), -131.9 (ddd, 1 F, $J_{F(2')-F(3')}$ = 21Hz, $J_{F(2')-F(5')}$ = 9 Hz, $J_{F(2')-F(4')}$ = 6 Hz, F-2′). GC–MS *m/z*: 470 (M^+), 280 (M^+ - NCC_6F_4O).

*4.2. Synthesis of tetraoxacalixarenes **3a,b – 6a,b**.*

*4.2.1. 5,6,16,17,26,28-Hexafluoro-2,8,14,20-tetraoxapentacyclo[19.3.1.1^{3,7}.1^{9,13}.1^{15,19}]octa-cosa-1(25),3(28),4,6,9(27),10,12,15(26),16,18,21,23-dodecaene-4,18-dicarbonitrile **3a** and 5,6,17,18,26,28-hexafluoro-2,8,14,20-tetraoxapentacyclo[19.3.1.1^{3,7}.1^{9,13}.1^{15,19}]octacosa-1(25),3(28),4,6,9(27),10,12,15(26),16,18,21,23-dodecaene-4,16-dicarbonitrile **3b***

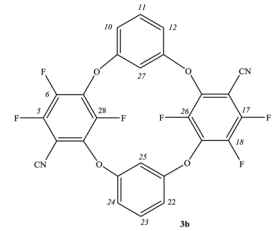


1 Solution of 0.22 g (2 mmol) of resorcinol and 0.77 g (~1.7 mmol) of triphenyl ethers **1a,b** (with 4% admixture of pentafluorobenzonitrile) in 50 mL of acetonitrile was added dropwise in period of 4 h with stirring into boiling solution of triethylamine (2.40 g, 24 mmol) in 50 mL of acetonitrile. The mixture was stirred under reflux for 23 h and the solvent was evaporated. Silica gel column chromatography (eluent $CCl_4/CHCl_3$ 3:1) gave 0.37 g of solid material containing (GC–MS, NMR 19F) 6% of triphenyl ethers **1a**, 84% and 5% of isomeric tetraoxacalixarenes **3a,b**, 2% of pentafluorobenzonitrile.

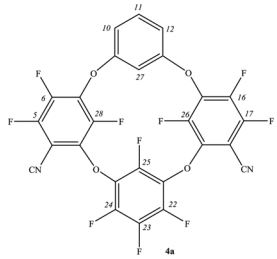
2 Solution of 0.22 g (2 mmol) of resorcinol and 0.77 g (4 mmol) of pentafluorobenzonitrile in 50 mL of acetonitrile was added dropwise in period of 2 h with stirring into solution of triethylamine (2.40 g, 24 mmol) in 50 mL of acetonitrile, cooled to 0 °C. The mixture was stirred for 4 h at room temperature and 10 min under reflux then a solution of resorcinol (0.22 g, 2 mmol) in 50 mL of acetonitrile was added dropwise with stirring for 2 h. The mixture was refluxed for 20 h and the solvent was evaporated. Silica gel column chromatography (eluent $CCl_4/CHCl_3$ 3:1) gave 0.60 g of solid material containing (GC–MS) 28% of triphenyl ethers **1a** (M^+ 456) and 70% of isomeric tetraoxacalixarenes **3a,b** (65% and 5%, M^+ 526), double-play crystallisation of which from CCl_4 gave 0.33 g (31%) of pure tetraoxacalixarene **3a**.

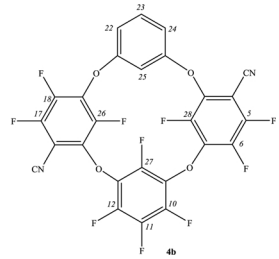
Tetraoxacalixarene **3a:** White powder, mp 299–302 °C (sealed capillary, with decomposition). IR (KBr) ν, cm^{-1}: 3063 (C_{Ar}-H); 2243 (-CN); 1421, 1446, 1489, 1601, 1637 (Ar, Ar_F); 1250, 1169, 1149, 1124 (Ar-O); 1007, 997, 960 (C–F). ^{19}F NMR ($CDCl_3$): δ -152.7 (d, 2 F, $J_{F(6)-F(5)}$ = 22 Hz, F-6,16), -137.8 (d, 2 F, $J_{F(28)-F(5)}$ = 11 Hz, F-26,28), -131.5 (dd, 2 F, $J_{F(5)-F(6)}$ = 22 Hz, $J_{F(5)-F(28)}$ = 11 Hz, F-5,17). ^{1}H NMR ($CDCl_3$): δ 5.95 (m, 1 H), 6.01 (m, 1 H) (H-25, H-27), 7.02 (dd, 2 H), 7.05 (dd, 2 H) ($J_{H(10)-H(11)}$ = 8 Hz, $J_{H(10)-H(27)}$ = 2 Hz, H-10,12, H-22,24), 7.32 (t, 1 H), 7.41 (t, 1 H) ($J_{H(11)-H(10,12)}$ = 8 Hz, H-11, H-23). ^{13}C NMR ($CDCl_3$): δ 94.67 (d, $^2J_{CF}$ = 14.8 Hz, $\underline{C}_4$-CN, $\underline{C}_{18}$-CN), 98.96, 99.22 ($\underline{C}_{25}$-H, $\underline{C}_{27}$-H), 108.80 ($\underline{C}_{10}$-H, $\underline{C}_{12}$-H, $\underline{C}_{22}$-H, $\underline{C}_{24}$-H), 113.62 (d, $^3J_{CF}$ = 13.6 Hz, 2$\underline{C}$N), 131.69 ($\underline{C}_{11}$-H, $\underline{C}_{23}$-H), 138.12 (t, $^2J_{CF}$ = 12.8 Hz, $\underline{C}_7$-O, $\underline{C}_{15}$-O), 140.87 (d, $^2J_{CF}$ = 12.0 Hz, $\underline{C}_3$-O, $\underline{C}_{19}$-O), 141.77 (ddd, $^1J_{CF}$ = 255.2 Hz, $^2J_{CF}$ = 13.6 Hz, J_{CF} = 3.7 Hz, $\underline{C}_5$-F, $\underline{C}_{17}$-F or $\underline{C}_6$-F, $\underline{C}_{16}$-F), 144.02 (dd, $^1J_{CF}$ = 258.5 Hz, J_{CF} = 4.4 Hz, $\underline{C}_{26}$-F, $\underline{C}_{28}$-F), 149.12 (ddd, $^1J_{CF}$ = 261.7 Hz, $^2J_{CF}$ = 13.2 Hz, J_{CF} = 3.2 Hz, $\underline{C}_6$-F, $\underline{C}_{16}$-F or $\underline{C}_5$-F, $\underline{C}_{17}$-F), 157.13, 157.14 ($\underline{C}_1$-O, $\underline{C}_9$-O, $\underline{C}_{13}$-O, $\underline{C}_{21}$-O). HRMS *m/z*: 526.0376 (M^+), 456, 263 (M^{++}). Calcd. for $C_{26}H_8F_6N_2O_4$ = 526.0383.

Tetraoxacalixarene **3b:** ^{19}F NMR ($CDCl_3$) from spectra of mixture **3a,b**: δ -152.4 (d, 2 F, $J_{F(6)-F(5)}$ = 22 Hz, F-6,18), -137.9 (d, 2 F, $J_{F(28)-F(5)}$ = 11 Hz, F-26,28), -131.4 (dd, 2 F, $J_{F(5)-F(6)}$ = 22 Hz, $J_{F(5)-F(28)}$ = 11 Hz, F-5,17). GC–MS *m/z*: 526 (M^+), 263 (M^{++}).



*4.2.2. 5,6,16,17,22,23,24,25,26,28-Decafluoro-2,8,14,20-tetraoxapentacyclo[19.3.1.1^{3,7}.1^{9,13}.1^{15,19}]octacosa-1(25),3(28),4,6,9(27),10,12,15(26),16,18,21,23-dodecaene-4,18-dicarbonitrile **4a** and 5,6,10,11,12,17,18,26,27,28-decafluoro-2,8,14,20-tetraoxapentacyclo[19.3.1.1^{3,7}.1^{9,13}.1^{15,19}]octacosa-1(25),3(28),4,6,9(27),10,12,15(26),16,18,21,23-dodecaene-4,16-dicarbonitrile **4b***





Solution of 0.22 g (2 mmol) of resorcinol and 0.77 g (4 mmol) of pentafluorobenzonitrile in 60 mL of acetonitrile was added dropwise in period of 7 h with stirring into solution of triethylamine (1.20 g, 12 mmol) in 20 mL of acetonitrile, cooled to 0 °C. The mixture was stirred for 1 h at room temperature and 1 h under reflux. Tetrafluororesorcinol (0.20 g, 1.1 mmol) was added and the mixture was refluxed for 9 h, then another portion of tetrafluororesorcinol (0.20 g, 1.1 mmol) was added and the mixture was refluxed for 27 h. After evaporation of the solvent silica gel column chromatography (eluent $CCl_4/CHCl_3$ 3:1) gave 0.70 g of viscous material containing (GC–MS, NMR ^{19}F) 93% and 4% of isomeric tetraoxacalixarenes **4a,b** and 2% of triphenyl ether **1a**, double-play crystallisation of which from CCl_4 gave 0.50 g (42%) of pure tetraoxacalixarene **4a**.

Tetraoxacalixarene **4a:** White crystals, mp 160–230 °C (sealed capillary, softening with the formation of transparent very viscous glass-like film). IR (KBr) ν, cm^{-1}: 3081 (C_{Ar}-H); 2245 (-CN); 1417, 1497, 1514, 1595, 1639 (Ar, Ar_F); 1304, 1248, 1165, 1149 (Ar-O); 1026, 968, 954 (C–F), 787, 764. ^{19}F NMR ($CDCl_3$): δ -160.1 (td, 1 F, $J_{F(23)-F(22,24)}$ = 21 Hz, $J_{F(23)-F(25)}$ = 5 Hz, F-23), -154.4 (m, 1 F, F-25), -153.4 (dd, 2 F, $J_{F(22)-F(23)}$ = 21 Hz, $J_{F(22)-F(25)}$ = 3 Hz, F-22,24), -152.3 (d, 2 F, $J_{F(6)-F(5)}$ = 21 Hz, F-6,16), -145.1 (d, 2 F, $J_{F(28)-F(5)}$ = 10 Hz, F-26,28), -130.8 (dd, 2 F, $J_{F(5)-F(6)}$ = 21 Hz, $J_{F(5)-F(28)}$ = 10 Hz, F-5,17). ^{1}H NMR ($CDCl_3$): δ 5.95 (m, 1H, H-27), 7.06 (dd, 2H, $J_{H(10)-H(11)}$ = 8 Hz, $J_{H(10)-H(27)}$ = 2 Hz, H-10,12), 7.43 (t, 1H, $J_{H(11)-H(10,12)}$ = 8 Hz, H-11). HRMS m/z: 598.0009 (M^+), 571 (M^+ - HCN), 570, 552 (M^+ - HCN - F), 551, 542, 523. Calcd. for C ($C_{26}H_4F_{10}N_2O_4$ = 598.0006.

Crystal data for **4a*0.54(CCl_4)**: tetragonal system, a = 12.9909(6), b = 12.9909(6), c = 17.5694(9) Å, V = 2965.1(3) Å^3, space group $P4_3$, Z = 4, d_{calc} = 1.526 g/cm^3, μ = 0.331 mm^{-1}. Refinement parameters wR_2 = 0.2359, S = 0.934 for all 5278 reflections (R = 0.0829 for 3339 observed reflections $F > 4\sigma$).

Tetraoxacalixarene **4b:** ^{19}F NMR ($CDCl_3$) from spectra of mixture **4a,b**: δ -160.0 (td, 1 F, $J_{F(11)-F(10,12)}$ = 21 Hz, $J_{F(11)-F(27)}$ = 5 Hz, F-11), -154.0 (m, 1 F, F-27), -153.8 (1 F), -153.2 (1 F) (dd, J = 21 Hz, J = 3 Hz, F-10, F-12), -152.8 (1 F), -152.0(1 F) (d, J = 21 Hz, F-6, F-18), -145.0 (d, 2 F, J = 10 Hz, F-26,28), -131.0 (1 F), -130.7(1 F) (dd, 2 F, J = 21 Hz, J = 10 Hz, F-5,17). ^{1}H NMR ($CDCl_3$): δ 5.91 (m, 1H, H-25), 7.06 (2H, H-22,24), 7.43 (t, 1H, J = 8 Hz, H-23). GC-MS m/z: 598 (M^+), 299 (M^{++}).

*4.2.3. 5,6,16,17,26,28-hexafluoro-11,23-dimethyl-2,8,14,20-tetraoxapentacyclo [19.3.1.1^{3,7}.1^{9,13}.1^{15,19}]octacosa-1(25),3(28),4,6,9(27),10,12,15(26),16,18,21,23-dodecaene-4,18-dicarbonitrile **5a** and 5,6,17,18,26,28-hexafluoro-11,23-dimethyl-2,8,14,20-tetraoxapentacyclo [19.3.1.1^{3,7}.1^{9,13}.1^{15,19}]octacosa-1(25),3(28),4,6,9(27),10,12,15(26),16,18,21,23-dodecaene-4,16-dicarbonitrile **5b***

Solution of 0.50 g (3.5 mmol) of orcinol monohydrate and 1.67 g (3.55 mmol) of triphenyl ethers **2a,b** (24:1) in 15 mL of acetonitrile was added with stirring into boiling solution of triethylamine (2.00 g, 20 mmol) in 110 mL of acetonitrile. The mixture was stirred under reflux for 30 h and the solvent was evaporated. Silica gel column chromatography (eluent $CCl_4/CHCl_3$ 3:1) gave 0.80 g (41%) of solid product containing (GC–MS, NMR ^{19}F) 98% of compound **5a** and ≤1% compound **5b.**

Tetraoxacalixarene **5a:** White powder, mp 295–297 °C (sealed capillary, with decomposition). IR (KBr) ν, cm^{-1}: 3068 (C_{Ar}-H); 2956, 2926, 2854 (C–H); 2241 (-CN); 1421, 1491, 1591, 1626 (Ar, Ar_F); 1292, 1157, 1120 (Ar-O); 926–1036 (group of bands, C–F). ^{19}F NMR ($CDCl_3$): δ -153.4 (d, 2 F, $J_{F(6)-F(5)}$ = 22 Hz, F-6,16), -137.8 (d, 2 F, $J_{F(28)-F(5)}$ = 11 Hz, F-26,28), -132.2 (dd, 2 F, $J_{F(5)-F(6)}$ = 22 Hz, $J_{F(5)-F(28)}$ = 11 Hz, F-5,17). ^{1}H NMR ($CDCl_3$): δ 2.30 (s, 3H, CH_3), 2.36 (s, 3H, CH_3), 5.81 (m, 1 H), 5.96 (m, 1 H) (H-25, H-27), 6.80 (d, 2 H), 6.84(d, 2 H) ($J_{H(10)-H(27)}$ = 2 Hz, H-10,12, H-22,24). ^{13}C NMR ($CDCl_3$): δ 21.18, 21.25 (2$\underline{C}H_3$), 94.41 (d, $^2J_{CF}$ = 16.2 Hz, $\underline{C}_4$-CN, $\underline{C}_{18}$-CN), 96.25, 96.45 ($\underline{C}_{25}$-H, $\underline{C}_{27}$-H), 108.98 ($\underline{C}_{10}$-H, $\underline{C}_{12}$-H, $\underline{C}_{22}$-H, $\underline{C}_{24}$-H), 114.09 (d, $^3J_{CF}$ = 9.9 Hz, 2$\underline{C}$N), 138.26 (t, $^2J_{CF}$ = 12.7 Hz, $\underline{C}_7$-O, $\underline{C}_{15}$-O), 140.94 (d, $^2J_{CF}$ = 12.0 Hz, $\underline{C}_3$-O, $\underline{C}_{19}$-O), 141.68 (ddd, $^1J_{CF}$ = 256.2 Hz, $^2J_{CF}$ = 13.6 Hz, J_{CF} = 4.1 Hz, $\underline{C}_5$-F, $\underline{C}_{17}$-F or $\underline{C}_6$-F, $\underline{C}_{16}$-F), 142.49, 142.73 ($\underline{C}_{11}$-CH_3, $\underline{C}_{23}$-CH_3), 144.10 (dd, $^1J_{CF}$ = 258.1 Hz, J_{CF} = 3.7 Hz, $\underline{C}_{26}$-F, $\underline{C}_{28}$-F), 149.07 (ddd, $^1J_{CF}$ = 261.5 Hz, $^2J_{CF}$ = 13.2 Hz, J_{CF} = 3.5 Hz, $\underline{C}_6$-F, $\underline{C}_{16}$-F or $\underline{C}_5$-F, $\underline{C}_{17}$-F), 156.93, 156.97 ($\underline{C}_1$-O, $\underline{C}_9$-O, $\underline{C}_{13}$-O, $\underline{C}_{21}$-O). HRMS m/z: 554.0693 (M^+), 298, 277 (M^{++}). Calcd. for $C_{28}H_{12}F_6N_2O_4$ = 554.0696.

*4.2.4. 5,6,16,17,22,23,24,25,26,28-decafluoro-11-methyl-2,8,14,20-tetraoxapentacyclo [19.3.1.1^{3,7}.1^{9,13}.1^{15,19}]octacosa-1(25),3(28),4,6,9(27),10,12,15(26),16,18,21,23-dodecaene-4,18-dicarbonitrile **6a** and 5,6,10,11,12,17,18,26,27,28-decafluoro-23-methyl-2,8,14,20-tetraoxapentacyclo[19.3.1.1^{3,7}.1^{9,13}.1^{15,19}]octacosa-1(25),3(28),4,6,9(27),10,12,15(26),16,18,21, 23-dodecaene-4,16-dicarbonitrile **6b***

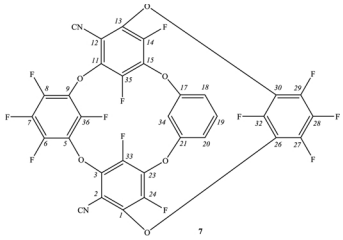
Solution of 0.28 g (2 mmol) of orcinol monohydrate and 0.77 g (4 mmol) of pentafluorobenzonitrile in 2 mL of acetonitrile was added with stirring into solution of triethylamine (2.40 g, 24 mmol) in 8 mL of acetonitrile. The mixture was stirred for 1.5 h at room temperature and 0.5 h at 70 °C. Tetrafluororesorcinol (0.36 g, 2 mmol) was added and the mixture was refluxed for 7.5 h. After evaporation of the solvent silica gel column chromatography (eluent $CCl_4/CHCl_3$ 3:1) gave 0.83 g of viscous material containing (GC–MS) 37% of triphenyl ethers **2a**, 11% and 51% of isomeric tetraoxacalixarenes **6a,b.** Crystallization of viscous material from petroleum ether (bp 40–70 °C) and then from CCl_4 gave 0.39 g (32%) of product containing (NMR ^{19}F) 93% of compound **6a**, 5% compound **6b** and 2% of triphenyl ethers **2a**.

Tetraoxacalixarene **6a:** White powder. IR (KBr) ν, cm^{-1}: 3066 (C_{Ar}-H); 1416, 1595,1514, 1585, 1626 (Ar, Ar_F); 1294, 1155 (Ar-O); 1024, 985, 927 (C–F). ^{19}F NMR ($CDCl_3$): δ -160.2 (td, 1 F, $J_{F(23)-F(22,24)}$ = 22 Hz, $J_{F(23)-F(25)}$ = 5 Hz, F-23), -154.0 (m, 1 F, F-25), -153.5 (dd, 2 F, $J_{F(22)-F(23)}$ = 22 Hz, $J_{F(22)-F(25)}$ = 2 Hz, F-22,24), -152.4 (d, 2 F, $J_{F(6)-F(5)}$ = 21 Hz, F-6,16), -145.0 (d, 2 F, $J_{F(28)-F(5)}$ = 10 Hz, F-26,28), -130.9 (dd, 2 F, $J_{F(5)-F(6)}$ = 21 Hz, $J_{F(5)-F(28)}$ = 10 Hz, F-5,17). ^{1}H NMR ($CDCl_3$): δ 2.38 (s, 3H, CH_3), 5.72 (m, 1H, H-27), 6.88 (d, 2H, $J_{H(10)-H(27)}$ = 2 Hz, H-10,12). HRMS m/z, 612.0165 (M^+). Calcd. for $C_{27}H_6F_{10}N_2O_4$ = 612.0162.

Tetraoxacalixarene **6b:** ^{19}F NMR ($CDCl_3$) from spectra of mixture **6a,b**: δ -160.1 (td, 1 F, $J_{F(11)-F(10,12)}$ = 21 Hz, $J_{F(11)-F(27)}$ = 5 Hz, F-11), -154.4 (m, 1 F, F-27), -153.8 (1 F), -153.3 (1 F) (dd, J = 21 Hz, J = 2 Hz, F-10, F-12), -152.9 (1 F), -152.0 (1 F) (d, J = 21 Hz, F-6, F-18), -144.9 (d, 2 F, J = 10 Hz, F-26,28), -131.0 (1 F), -130.8 (1 F) (dd, 2 F, J = 21 Hz, J = 10 Hz, F-5,17). ^{1}H NMR ($CDCl_3$): δ 2.38 (3H, CH_3), 5.68 (m, 1H, H-25), 6.88 (d, 2H, $J_{H(10)-H(27)}$ = 2 Hz, H-22,24). GC–MS m/z: 612 (M^+), 306 (M^{++}).

4.3. Synthesis of bicyclooxacalixarenes 7 and 8

*4.3.1. 6,7,8,14,24,27,28,29,32,33,35,36-dodecafluoro-4,10,16,22,25,31-hexaoxaheptacyclo [11.11.7.1^{3,23}.1^{5,9}.1^{11,15}.1^{17,21}.1^{26,30}]hexatriaconta-1,3(33),5(36),6,8,11(35),12,14,17(34),18,20, 23,26(32),27,29-pentadecaene-2,12-dicarbonitrile **7***



7

Triethylamine (1.20 g, 12 mmol) was added to solution of tetrafluororesorcinol (0.36 g, 2 mmol) and 0.82 g of mixture containing (GC–MS) 25% of triphenyl ether **1a** and isomeric tetraoxacalixarenes **4a,b** (69% and 4%) in 20 mL of acetonitrile. The mixture was stirred under reflux for 40 h and the solvent was evaporated. Silica gel column chromatography (eluent $CCl_4/CHCl_3$ 3:1) gave 0.40 g of solid product containing (GC-MC) isomeric tetraoxacalixarenes **4a,b** (79% and 3%), 14% of bicyclooxacalixarene **7** and 0.18 g of viscous material containing (GC–MS) isomeric tetraoxacalixarenes **4a,b** (73% and 3%), 23% of bicyclooxacalixarene **7**. Crystallization of combined material from CH_2Cl_2 and then from benzene gave ~0.09 g of complex **7*3C$_6$H$_6$**.

Bicyclooxacalixarene **7**: ^{19}F NMR ($CDCl_3$): δ -159.5 (tt, 2 F, $J_{F(7)-F(6,8)}$ = 21 Hz, $J_{F(7)-F(36)}$ = 2 Hz, F-7,28), -151.4 (m, 2 F, F-32,36), -150.3 (d, 4 F, $J_{F(6)-F(7)}$ = 21 Hz, F-6,8,27,29), -138.7 (t, 4 F, $J_{F(14)-F(32,36)}$ ~4 Hz, F-14,35,24,33). ^{1}H NMR ($CDCl_3$): δ 4.76 (m, 1H, H-34), 7.07 (dd, 2H, $J_{H(18)-H(19)}$ = 8 Hz, $J_{H(18)-H34)}$ = 2 Hz, H-18,20), 7.45 (t, 1H, $J_{H(19)-H(18)}$ = 8 Hz, H-19). HRMS m/z, 739.9869 (M^+). Calcd. for $C_{32}H_4F_{12}N_2O_6$ = 739.9872.

Crystal data for **7*3C$_6$H$_6$**: monoclinic system, a = 11.696(3), b = 15.442(3), c = 21.170(4) Å, β = 99.640(10)°, V = 3769.6(14) Å^3, space group $C2/m$, Z = 4, d_{calc} = 1.580 g/cm^3, μ = 0.145 mm^{-1}. Refinement parameters wR_2 = 0.1548, S = 1.101 for all 3908 reflections (R = 0.0457 for 2820 observed reflections $F > 4\sigma$).

*4.3.2. 14,24,33,35-tetrafluoro-7,19,28-trimethyl-4,10,16,22,25,31-hexaoxaheptacyclo [11.11.7.1^{3,23}.1^{5,9}.1^{11,15}.1^{17,21}.1^{26,30}]hexatriaconta-1,3(33),5(36),6,8,11(35),12,14,17(34),18,20, 23,26(32),27,29-pentadecaene-2,12-dicarbonitrile **8**.*

Triethylamine (1.20 g, 12 mmol) was added to solution of orcinol (0.28 g, 2 mmol) and 0.90 g of mixture containing (GC–MS) 10% of triphenyl ether **2a** and isomeric tetraoxacalixarenes **5a,b** (80% and 9%) in 150 mL of acetonitrile. The mixture was stirred under reflux for 27 h and the solvent was evaporated. The reaction mixture was treated with 50 mL of 5% aqueous HCl and then with methylene chloride (3 × 50 mL). The extract was dried over Na_2SO_4 and evaporation gave 1.12 g of a viscous material. Silica gel column chromatography (eluent $CCl_4/CHCl_3$ 3:1) gave 0.47 g of solid product containing (GC-MC) isomeric tetraoxacalixarenes **5a,b** (M^+ 554, 68% and 3%) and 27% of bicyclooxacalixarene **8** (M^+ 638). By triple crystallization of product from mixture (4:1) of $CHCl_3$ and petroleum ether (bp 40–70 °C) 0.12 g of product containing (GC-MC) of isomeric tetraoxacalixarenes **5a,b** (90% and 5%) and 4% of bicyclooxacalixarene **8** was obtained. Additionally crystallization of mother liquor (0.35 g) from mixture (3:1) of CH_2Cl_2 and petroleum ether (bp 40–70 °C) gave 0.15 g of solid product containing (^{19}F NMR) bicyclooxacalixarene **8** and isomeric tetraoxacalixarenes **5a** in 69:31 ratio.

Bicyclooxacalixarene **8**: ^{19}F NMR ($CDCl_3$): δ -140.1 (s, 4 F, F-14,24,33,35). ^{1}H NMR ($CDCl_3$): δ 2.39 (s, 3H, CH_3), 2.41 (s, 6H, 2CH_3), 4.87 (t, 2H, $J_{H(36)-H(6,8)}$ = 2.2 Hz, H-32,36), 5.02 (m, 1H, H-34), 6.88 (dd, 2H, $J_{H(18)-H(34)}$ = 2.2 Hz, J ≈ 0.5 Hz, H-18,20), 6.91(dd, 4H, $J_{H(6)-H(36)}$ = 2.2 Hz, J ≈ 0.5 Hz, H-6,8, H-27,29). HRMS m/z: 638.1099 (M^+), 554, 319 (M^{++}). Calcd. for $C_{35}H_{18}F_4N_2O_6$ = 638.1096.

4.4. Complexation of oxacalixarenes 5a, 8 and 7 with $NiCl_2$ and $AgNO_3$ salts

1. $NiCl_2$*6H_2O (0.0024 g) and mixture of bicyclooxacalixarene **8** with tetraoxacalixarene **5a** (0.0064 g) was dissolved with heating in 8 ml of methanol. The solution was kept at 20 °C for 2 days and the spectrum of ESI-MS (+) was recorded. Figure (Fig. 4a) and analysis of the spectrum are given in the **Results and discussion** part. In addition to the complex ions listed in the text, there are also [M$_8$ + Na]$^+$ (m/z 661) and [2 M$_8$ + Na]$^+$ (m/z 1299) ions in the spectrum.

2. $AgNO_3$ (0.0018 g) and mixture of bicyclooxacalixarene **8** with tetraoxacalixarene **5a** (0.0064 g) was dissolved with heating in 10 ml of methanol. The solution was kept at 20 °C for 2 days and the spectrum of ESI-MS (+) was recorded. Figure (Fig. 4b) and analysis of the spectrum are given in the **Results and discussion** part. In addition to the complex ions listed in the text, there are also [M$_8$ + H + CH_3OH]$^+$ (m/z 671) ion in the spectrum.

3. Bicyclooxacalixarene **7** (0.0019 g) and 0.0011 $NiCl_2$*6H_2O was dissolved with heating in 8 ml of methanol. The solution was kept at 20 °C for 2 days and the spectrum of ESI-MS (+) was recorded. In spectrum there is only ion [M$_7$ + Na]$^+$ (m/z 763).

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